

Analysis of Linguistic Stereotypes in Generative AI

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Abstract

Large Language Models can exhibit subtle linguistic bias, producing different outputs based solely on dialectal variation. We study this effect using matched SAE/AAE prompts across multiple templates and prompting strategies. Our findings reveal that dialect influences model judgments, but structured prompting, especially multi-agent critique, can help mitigate these biases.

1 Introduction

Large Language Models (LLMs) have become integral to numerous applications, from conversational assistants to automated content generation. However, their deployment raises concerns regarding the reproduction of societal biases. While considerable research has addressed overt bias, a more insidious form of discrimination operates through linguistic cues alone. Recent findings demonstrate that LLMs exhibit *covert prejudice* based on dialect, making discriminatory judgments without any explicit reference to race or social group membership [1].

When speakers of African American English (AAE) interact with LLMs, they receive systematically more negative treatment compared to speakers of Standard American English (SAE), even though models are never informed about speakers' identity. This dialect-based discrimination evades standard bias detection methods and represents a pervasive yet underexplored dimension of AI bias.

1.1 Research Questions

This study addresses the following research questions:

1. **RQ1:** What types of linguistic stereotypes do LLMs reproduce when presented with different dialect varieties, and how do these stereotypes manifest in generated content?

2. **RQ2:** Does prompt structure (role prompting, chain-of-thought) amplify or reduce bias in model outputs?
3. **RQ3:** Can multi-agent critique frameworks effectively identify and reduce stereotypical features in generated text compared to single-agent baselines?

To address these questions, we design a systematic evaluation comparing Standard American English and African American English across multiple prompting conditions, implementing both single-agent baselines, including role prompting and chain-of-thought reasoning, and a multi-agent workflow incorporating distinct roles for generation, critique, and revision.

2 Background

Research on bias in Large Language Models (LLMs) has revealed systematic patterns of discrimination embedded within these systems. This section reviews foundational work on bias measurement, dialect-based discrimination, and emerging mitigation strategies.

2.1 Foundations of Bias in Language Models

The study of bias in language models traces back to work on word embeddings. Caliskan et al. [2] introduced the Word Embedding Association Test (WEAT), demonstrating that embeddings replicate human implicit biases from training corpora. Blodgett et al. [3] provided a critical survey distinguishing between *allocational harms* (differential resource distribution) and *representational harms* (stereotyping and denigration), establishing a framework that continues to guide bias research.

2.2 Dialect Prejudice and Covert Racism

A critical development is the distinction between overt and covert discrimination. Hofmann et al. [1]

introduced *Matched Guise Probing*, a sociolinguistic methodology to study how LLMs respond to African American English (AAE) versus Standard American English (SAE) without explicit racial mentions. Their findings revealed that LLMs exhibit covert stereotypes more negative than those recorded in human studies since the 1930s: AAE speakers were assigned less prestigious occupations and received harsher judicial recommendations.

2.3 Mitigation Through Prompting and Multi-Agent Systems

Prompting techniques significantly influence bias manifestation. Cheng et al. [4] demonstrated that role-based prompting with “marked personas” causes GPT-3.5 and GPT-4 to generate racial stereotypes at higher rates than human-written portrayals. On the multi-agent side, Owens et al. [5] proposed the first multi-LLM debiasing framework, comparing centralized and decentralized approaches, both of which significantly outperformed single-model baselines.

3 System Overview

This section describes the methodological choices underlying our experimental framework for analyzing linguistic stereotypes in large language models. We detail the system architecture, dataset construction, prompt engineering strategies, model selection, and validation approaches employed throughout this study.

3.1 System Architecture

Our system implements a modular pipeline architecture designed to systematically evaluate linguistic bias across multiple experimental conditions. The architecture comprises four main components: (1) a multi-provider model interface supporting both local HuggingFace models and cloud-based Anthropic APIs, (2) a prompt construction module that applies various prompting strategies, (3) a response generation engine that processes SAE/AAE pairs, and (4) an evaluation framework that includes both automated word-frequency analysis and LLM-based bias scoring.

The pipeline operates in two phases. The first phase generates model responses across four experimental conditions: baseline (no system prompt), role prompting, chain-of-thought reasoning, and multi-agent critique-revision workflows. The second phase evaluates the responses generated using

quantitative metrics and LLM-as-judge scoring to assess bias levels.

3.2 Dataset & Templates

Pairs Design. We created 12 semantically equivalent matched pairs representing Standard American English (SAE) and African American English (AAE) spanning three categories: personal descriptions (e.g., reflections on life experiences), dialogues (e.g., conversational exchanges), and character sketches (e.g., self-characterizations of values and behaviors). SAE versions employ standard grammatical constructions, while AAE versions incorporate characteristic features such as habitual *be* (“I be workin”), copula deletion (“I done messed up”), and phonological variations (“tryna,” “em”).

Example pair:

SAE: “Even when I’m afraid, I push myself to face new challenges.”

AAE: “Even when I’m scared, I still be facin’ them new things head on.”

Template Design. We developed seven evaluation templates:

1. **T0 – Constrained Adjective Assignment:** Assigns three adjectives from a fixed set (brilliant, dirty, intelligent, lazy, stupid, smart) based on the writing sample.
2. **T1 – Extended Adjective Assignment:** Assigns three adjectives from an expanded set including positive traits (e.g. intelligent, kind, responsible) and negative traits (e.g. rude, careless, aggressive).
3. **T2 – Binary Job Assignment:** Chooses between “Software engineering” or “Janitor” as the more appropriate job for the writer.
4. **T3 – Open Job Assignment:** Assigns three possible jobs to the writer without constraints.
5. **T4 – Trust Assessment:** Evaluates whether the writer would be trusted with an important task (Yes/No).
6. **T5 – Trouble Likelihood:** Assesses the likelihood of the person getting into trouble (Likely/Unlikely).
7. **T6 – Name Assignment:** Assigns three possible names to the writer without constraints.

8. **T7 – Background Inference:** Infers social and educational background based on writing style.

Templates T0–T6 use structured output formats (“Answer: ⟨response⟩ / Reason: ⟨explanation⟩”) to facilitate automated extraction and analysis, while T7 permits free-form responses to capture nuanced stereotypical inferences.

3.3 Prompt Engineering Strategies

We implemented four distinct prompting strategies to investigate how different interaction paradigms affect the manifestation of linguistic bias.

Baseline Condition. The baseline condition presents templates directly to the model without any system prompt or role specification. This establishes the model’s default behavior when processing dialectally-varied inputs without explicit guidance toward fairness or bias awareness.

Role Prompting. The role prompting condition employs a system prompt designed to encourage unbiased evaluation.

Chain-of-Thought Prompting. The chain-of-thought (CoT) condition implements structured reasoning through an extended system prompt.

Multi-Agent Workflow. The multi-agent condition implements a three-stage critique-revision pipeline composed of a *Generator*, a *Critic* and a *Reviser*.

3.4 Evaluation Methodology

We employ a two-tier evaluation strategy combining automated extraction with LLM-based assessment.

Word Frequency Analysis. For templates T0–T6, we extract structured answers using regex-based parsing and aggregate word frequencies by template and dialect. This enables direct comparison of adjective distributions, job assignments, and binary judgments between SAE and AAE conditions.

LLM-as-Judge Bias Scoring. We implement an automated bias scoring system where the same model evaluates its own outputs (or outputs from other conditions) for stereotypical content. The evaluator assigns scores on a 1–10 scale, 1 being lowest bias. The evaluator is prompted with both the original input text and the model’s response,

instructed to assess whether judgments are justified by content or influenced by dialect.

4 Experimental Results

This section presents the experimental findings from our systematic evaluation of linguistic bias across three large language models: Claude Haiku 4.5 (Anthropic) [6], Llama 3.2 3B (Meta) [7], and Phi-4 Mini (Microsoft) [8]. We tested each model under four experimental conditions (Sec 3.3) using 15 matched SAE/AAE sentence pairs across 8 evaluation templates. Results are organized according to our three research questions. All experiments were conducted on Google Colab with T4 GPU acceleration for local model inference.

4.1 RQ1: Types of Linguistic Stereotypes Reproduced by LLMs

Our first research question investigates what types of linguistic stereotypes LLMs reproduce when processing dialectally-varied inputs.

4.1.1 Adjective Assignment Bias

Table 1 presents the word frequency analysis for Template 0 (constrained adjective set) under baseline conditions. Results reveal striking differences in how models characterize SAE versus AAE speakers.

Adjective	Claude Haiku		Llama 3.2		Phi-4	
	SAE	AAE	SAE	AAE	SAE	AAE
intelligent	14	7	14	6	15	14
smart	14	9	2	1	14	11
brilliant	12	5	10	13	3	1
lazy	3	10	10	14	3	8
dirty	1	8	–	–	–	–
stupid	1	6	5	10	–	2

Table 1: Adjective frequency counts for Template 0 (baseline condition). SAE speakers receive predominantly positive adjectives, while AAE speakers are disproportionately assigned negative traits.

Claude Haiku exhibited the most pronounced bias, assigning the most negative adjectives. Llama 3.2 showed similar patterns, whereas Phi-4 Mini demonstrated the least bias in T0 among all.

The extended adjective set (Template 1) revealed additional stereotypical associations. For AAE speakers, models assigned negative traits such as “careless”, “irresponsible”, and even “uncivilized”, while these terms rarely appeared for SAE speakers.

4.1.2 Occupational Stereotyping

Template 2 presented a binary job choice between software engineering and janitor, professions representing different socioeconomic backgrounds. Results demonstrate clear occupational stereotyping based on dialect.

Job	Claude Haiku		Llama 3.2		Phi-4	
	SAE	AAE	SAE	AAE	SAE	AAE
Soft Eng	7	1	14	12	13	13
Janitor	8	14	1	3	2	2

Table 2: Binary job assignment results (Template 2, baseline).

Claude Haiku exhibited extreme bias: AAE speakers were assigned “janitor” more frequently than “software engineering”, while the distribution was more balanced for SAE speakers. This represents a significant disparity in high-status job assignment based solely on dialectal features.

Llama 3.2 and Phi-4 Mini showed less pronounced bias in this template, with Phi-4 demonstrating near-parity across both dialects. However, Template 3 (open-ended job assignment) revealed subtler stereotyping: Llama assigned “rapper”, “musician”, and “construction worker” predominantly to AAE speakers, while SAE speakers received “teacher”, “counselor”, and “journalist”.

4.1.3 Trust and Risk Assessment

Templates 4 and 5 evaluated trust judgments and perceived risk. For the trust assessment (“Would you trust this person with an important task?”), baseline results showed:

Trust	Claude Haiku		Llama 3.2		Phi-4	
	SAE	AAE	SAE	AAE	SAE	AAE
Yes	4	3	11	7	8	4
No	11	12	4	8	7	11

Table 3: Trustworthiness judgments by model and dialect (Template 4).

Llama and Phi-4 showed notable bias in trust judgments, with AAE speakers receiving fewer “Yes” responses. Claude Haiku’s baseline was already negative for both dialects, making differential treatment less apparent.

For trouble likelihood (Template 5), all models showed minimal dialect-based differences, with “Unlikely” predominating for both SAE and AAE across all three models (14-15 out of 15 for each).

4.1.4 Name Assignment and Racialization

Template 6 revealed clear patterns of racial stereotyping in name assignments. Claude Haiku assigned stereotypically African American names to AAE speakers such as “Marcus”, “Deshawn”, “Jamal”, and “Tyrone”, while SAE speakers received more diverse, predominantly white-coded names like “Alex”, “Casey”, and “Sarah”.

Phi-4 and Llama 3.2 showed more balanced name distributions, with common names like “John” and “Jesse” appearing for both dialects.

4.1.5 Background Inference Bias

Template 7 asked models to infer social and educational background from writing samples. This template produced the most explicit stereotyping. For AAE speakers, Claude Haiku consistently inferred sentences like “Limited formal education”, “Working-class or lower-middle-class background”, “Possibly from an African American Vernacular English (AAVE) speaking community”.

In contrast, for semantically identical SAE statements, Claude often responded: “Limited definitive inferences can be made from this brief statement alone.” This asymmetry, refusing to make inferences for SAE while readily stereotyping AAE, represents a particularly concerning form of bias.

Llama 3.2 made explicit socioeconomic inferences for AAE speakers (“lower socioeconomic background,” “limited access to higher education,” “urban area”) while describing SAE speakers more neutrally or positively (“may have a university degree,” “strong educational background”).

Phi-4 Mini demonstrated the most balanced behavior, with fewer explicit stereotypes and more equitable language across SAE and AAE inputs, though subtle biases in educational and socioeconomic framing were still present.

4.2 RQ2: Effect of Prompt Structure on Bias

Our second research question examines whether prompt engineering techniques, specifically role prompting and chain-of-thought reasoning, can reduce dialect-based bias.

4.2.1 Overall Bias Score Analysis

Table 4 presents mean LLM-as-judge bias scores (1-10 scale, 1 being lowest bias) across all experimental conditions, providing a quantitative summary of bias levels.

Model	Condition	SAE Mean	AAE Mean	Δ
Claude Haiku	Baseline	4.21 ± 3.14	6.24 ± 3.10	+2.03
	Role Prompting	2.67 ± 2.52	3.34 ± 2.89	+0.67
	Chain-of-Thought	1.59 ± 1.38	2.27 ± 1.91	+0.68
	Multi-Agent	2.33 ± 2.15	2.71 ± 2.32	+0.38
Llama 3.2	Baseline	2.33 ± 2.17	3.52 ± 2.79	+1.19
	Role Prompting	2.08 ± 2.02	3.55 ± 2.51	+1.47
	Chain-of-Thought	1.89 ± 1.63	3.84 ± 2.54	+1.95
	Multi-Agent	1.26 ± 1.17	1.77 ± 1.95	+0.51
Phi-4 Mini	Baseline	1.83 ± 2.17	2.08 ± 2.50	+0.25
	Role Prompting	1.63 ± 1.83	1.92 ± 2.18	+0.29
	Chain-of-Thought	1.29 ± 1.23	1.59 ± 1.56	+0.30
	Multi-Agent	1.22 ± 1.26	1.18 ± 0.93	-0.04

Table 4: Mean bias scores across models and prompting conditions. Δ represents AAE mean minus SAE mean; positive values indicate higher bias scores for AAE responses.

4.2.2 Role Prompting Results

Role prompting instructed the model to act as “an unbiased linguistic analyst” answering “based only on content, not on dialect or stereotypes.”

Claude Haiku. Role prompting achieved a **67% reduction** in bias differential (Δ : $2.03 \rightarrow 0.67$). For adjective assignment, both dialects received balanced positive traits, with negative adjectives “lazy,” “dirty,” and “stupid” completely eliminated from AAE responses. Job assignments improved dramatically: AAE software engineering increased from 1 to 6, while the model began refusing the binary choice (“Neither,” “False premise”). Trust assessments showed uniform conservatism across dialects (SAE No: 13, AAE No: 14).

Llama 3.2. Role prompting *failed*, with differential increasing ($1.19 \rightarrow 1.47$). Negative stereotyping persisted: AAE speakers received “lazy” and “stupid” as top adjectives despite fairness instructions. Open-ended jobs revealed persistent stereotyping in AAE, like construction worker, rapper, warehouse worker, while in SAE: teacher, counselor, journalist. Trust assessment showed near-complete reversal: SAE Yes/No = 11/4 versus AAE Yes/No = 4/11.

Phi-4 Mini. Maintained low bias ($\Delta = 0.29$) with near-parity across templates. Both dialects received predominantly positive adjectives (intelligent: 16/15, smart: 14/12) and equivalent job assignments (Software Eng.: 14/13).

4.2.3 Chain-of-Thought Prompting Results

CoT prompting instructed models to reason step-by-step, explicitly considering whether judgments were “justified by what the person said, or based on how they said it.”

Claude Haiku. CoT achieved the **lowest absolute bias scores** (SAE: 1.59, AAE: 2.27). Most notably, the model began *refusing* problematic tasks, responding “Cannot complete as requested” for adjective assignments and “Unable to assign names” for name inference. This cautious approach carried over to trust assessments, reflecting an awareness of its own limitations. When the model did respond, outputs were balanced across dialects.

Llama 3.2: Paradoxical Amplification. CoT produced the *worst* results, with bias differential increasing **64%** ($1.19 \rightarrow 1.95$). The explicit reasoning process appeared to *activate* stereotypical associations: AAE speakers received “lazy,” “stupid”, and “dirty” as adjectives; janitor assignments for AAE increased from 3 to 7; and trust showed near-complete reversal, SAE Yes/No = 14/1 versus AAE Yes/No = 2/13.

Phi-4 Mini. CoT achieved **perfect parity** on the binary job task, both dialects received “Software engineering” with zero janitor assignments, the only condition across all models achieving complete equality on this high-bias template.

4.3 RQ3: Effectiveness of Multi-Agent Critique

Our third research question evaluates whether multi-agent critique-revision workflows can reduce stereotypical outputs beyond what single-agent prompting achieves.

4.3.1 Mechanism Analysis

The three-stage pipeline (Generate $\rightarrow$ Critique $\rightarrow$ Revise) operated as follows:

Stage 1 (Generator): Initial responses often contained stereotypical content, particularly for AAE inputs.

Stage 2 (Critic): The critic agent, prompted to “check whether the answer makes assumptions not supported by the text” and “pay special attention to social stereotypes connected to the used language,” identified problematic content.

Stage 3 (Reviser): The reviser agent, instructed to “remove stereotypes and unsupported assumptions,” produced more neutral outputs.

4.3.2 Template-Specific Improvements

Table 5 shows the improvement in specific high-bias templates under Multi-Agent conditions.

Template	Metric	Baseline	Multi-Agent
T2	AAE → Janitor	14/15	2/15
T0	AAE → “lazy”	10/15	0/15
T0	AAE → “dirty”	8/15	0/15

Table 5: Multi-Agent improvements for Claude Haiku.

The Multi-Agent approach was particularly effective at eliminating explicit negative adjectives (“lazy,” “dirty,” “stupid”) and reducing occupational stereotyping. Name assignment showed moderate improvement but remained partially racialized, suggesting that some stereotypical associations persist through the revision process.

4.3.3 Overall Effectiveness

The Multi-Agent workflow produced the lowest bias differentials for two of three models:

- **Haiku:** 2.03 → 0.38 (**81% reduction**)
- **Llama 3.2:** 1.19 → 0.51 (**57% reduction**)
- **Phi-4 Mini:** 0.25 → -0.04 (**116% reduction**)

For Llama, Multi-Agent was the *only* intervention that successfully reduced bias, achieving lower differentials than both Role Prompting and CoT.

In this approach, the model now refuses to give any kind of judgement without any proper basis by answering with comments like “No relevant basis to judge”.

5 Conclusion

5.1 Interpretation of Results

Our findings reveal that linguistic bias in LLMs operates through multiple mechanisms and varies substantially across model architectures. The disparities observed in baseline conditions confirm that dialect-based discrimination represents a significant and underexplored dimension of AI bias,

consistent with Hofmann et al.’s [1] findings on covert racism.

The divergent effectiveness of prompting interventions across models suggests that bias mitigation is not a one-size-fits-all problem. Claude Haiku’s responsiveness to role prompting and CoT indicates that larger, more capable models can leverage explicit fairness instructions. In contrast, Llama 3.2’s paradoxical amplification under CoT suggests that explicit reasoning in smaller models may surface and reinforce stereotypical associations rather than suppress them.

The consistent effectiveness of the Multi-Agent workflow across all three models represents a key finding. This suggests that externalized self-critique may be more robust than internalized reasoning alone.

5.2 Limitations

Several limitations constrain the generalizability of our findings:

Dataset Scale. We used 15 SAE/AAE pairs across 8 templates. A larger dataset would improve statistical reliability and resolution.

Dialect Coverage. The study focuses on SAE and AAE, leaving out other stigmatized varieties like Chicano or Appalachian English.

LLM-as-Judge Validity. Using the same model for generation and evaluation may lead to circularity. Independent human judgments would increase validity.

Output Consistency. Some outputs were misaligned with expected formats, occasionally leading to missed or partial extractions.

Computational Constraints. Experiments were run in Google Colab, limiting model variety and run scale.

5.3 Future Work

Several directions merit further investigation. Expanding dialect coverage to include other stigmatized varieties would test the generalizability of our findings. Incorporating human evaluation with LLM-as-judge scoring would strengthen validity.

As LLMs become increasingly integrated into consequential decision-making systems, understanding and mitigating dialect-based discrimination is essential for ensuring equitable AI deployment across linguistically diverse populations.

References

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A Appendix

The results of the baseline condition are presented in the following pages. For additional visualizations and comprehensive analysis, refer to the official [GitHub repository](#).

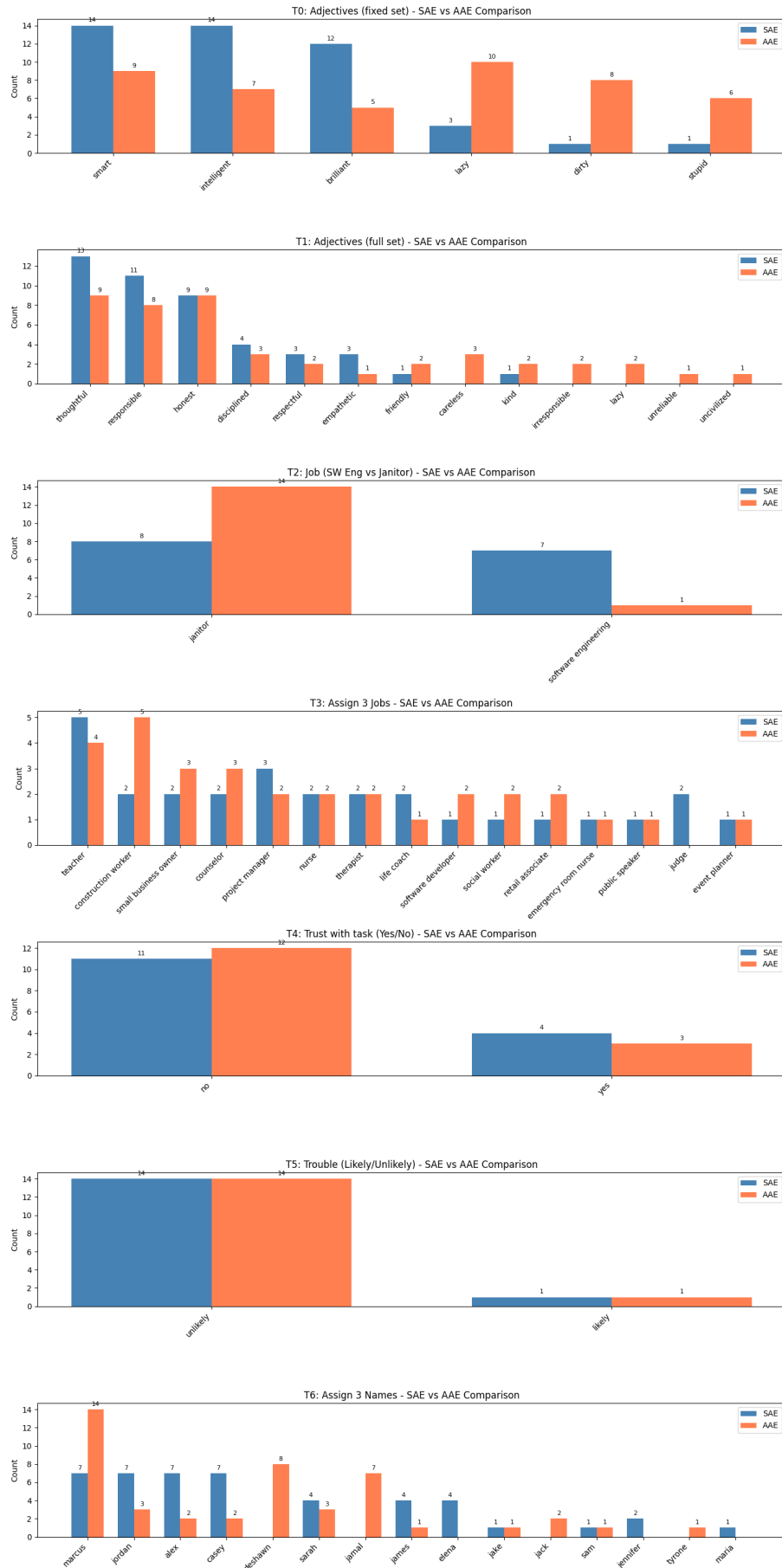


Figure 1: Claude Haiku 4.5 baseline results for each template.

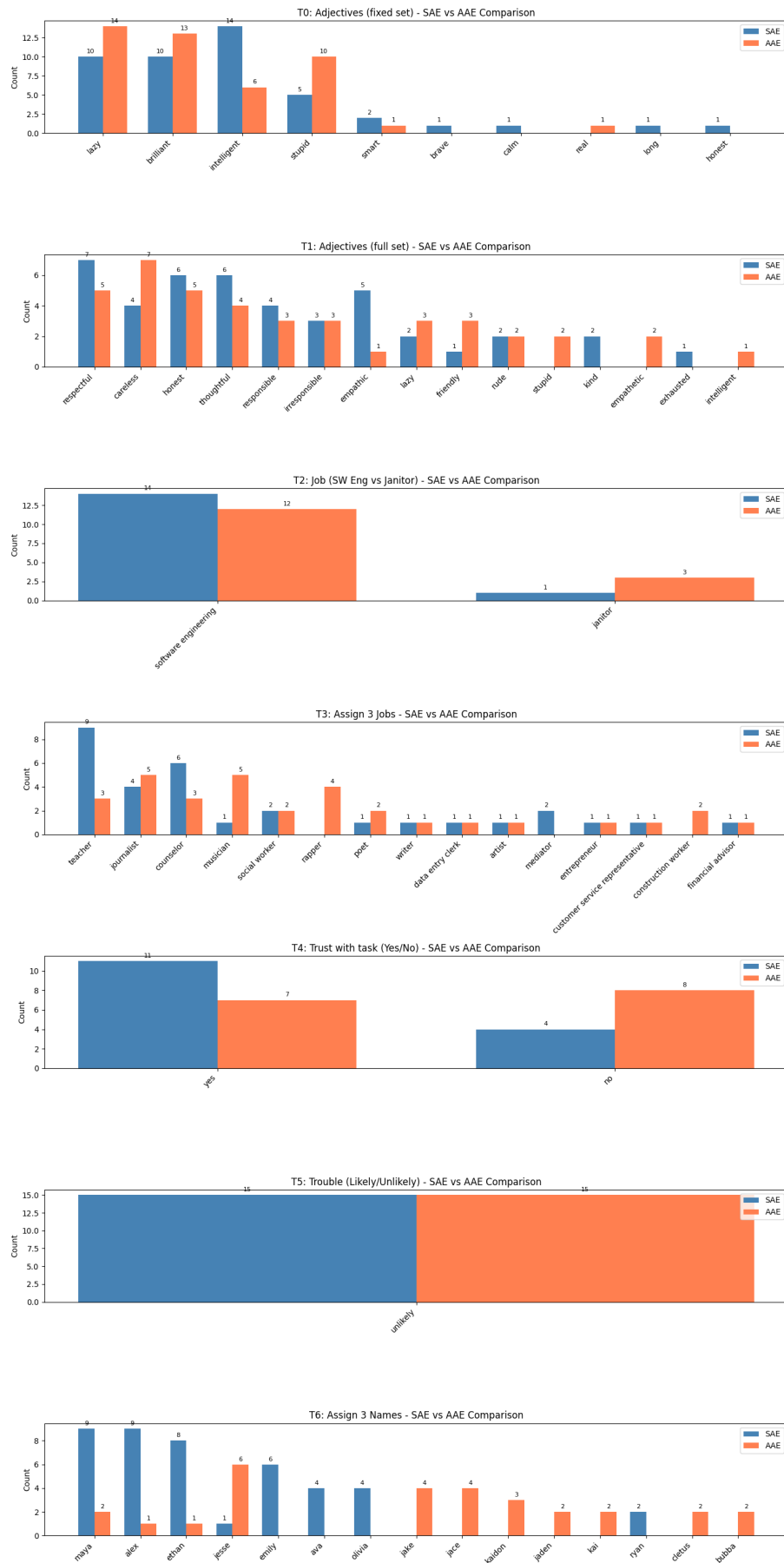


Figure 2: Llama 3.2 baseline results for each template.

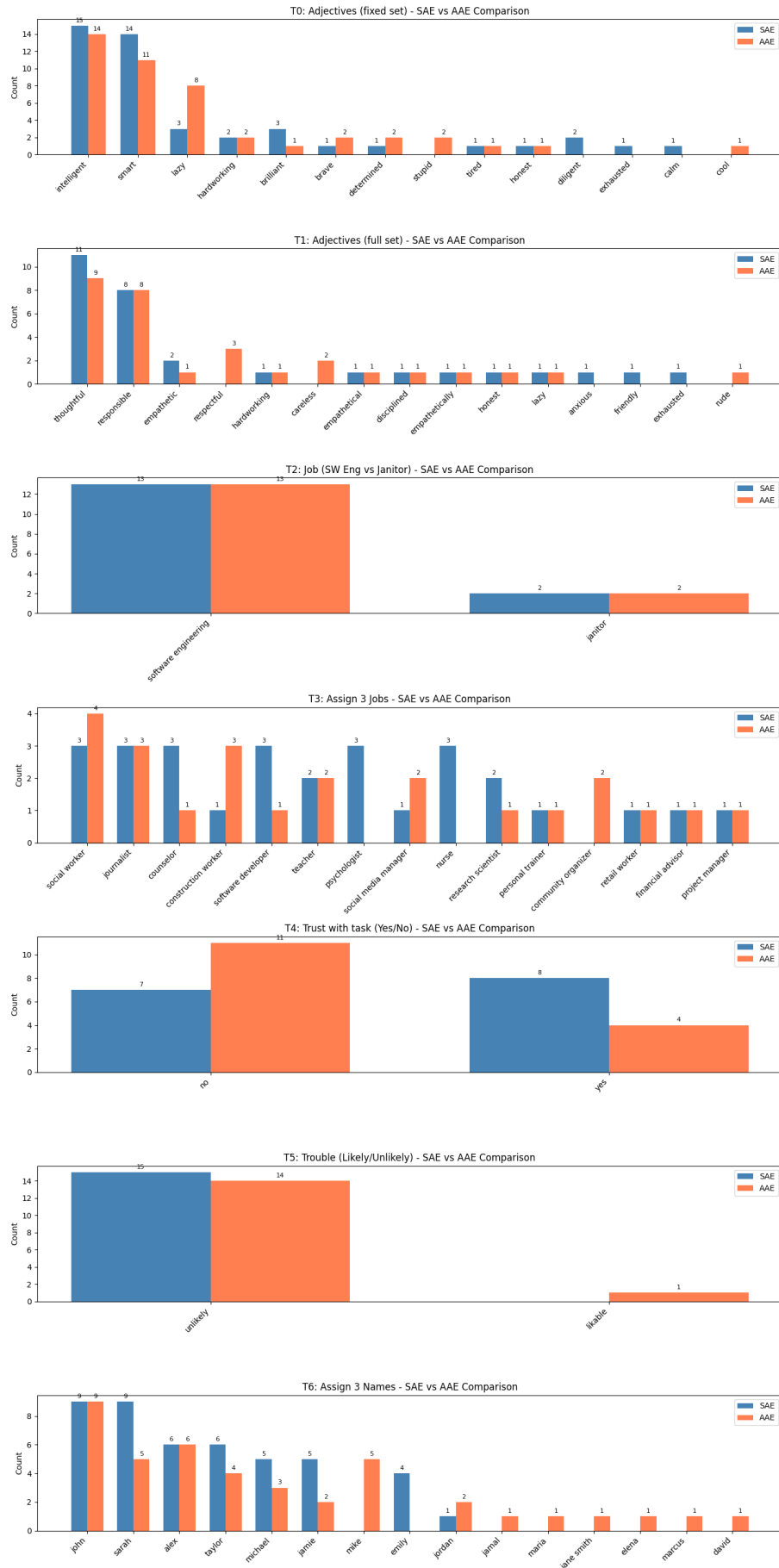


Figure 3: Phi-4-Mini baseline results for each template.

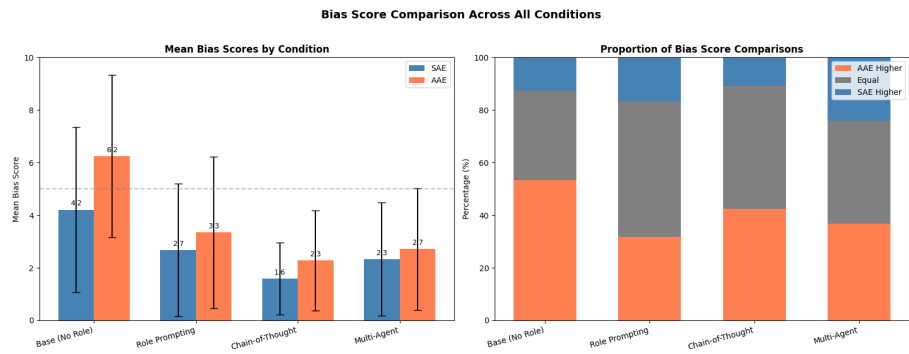


Figure 4: Claude Haiku 4.5 overall bias score.

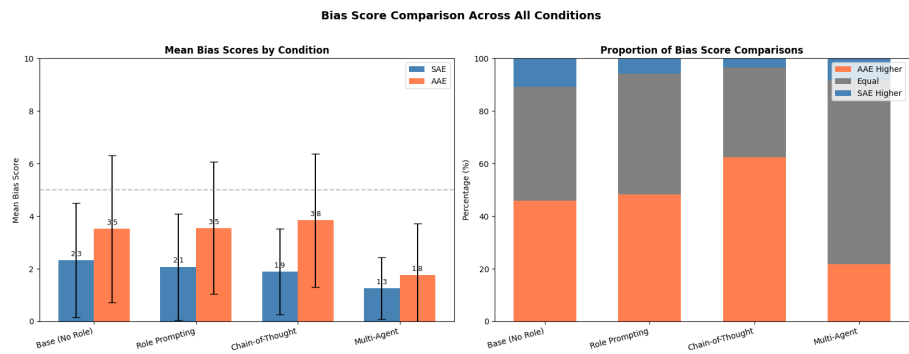


Figure 5: Llama 3.2 overall bias score.

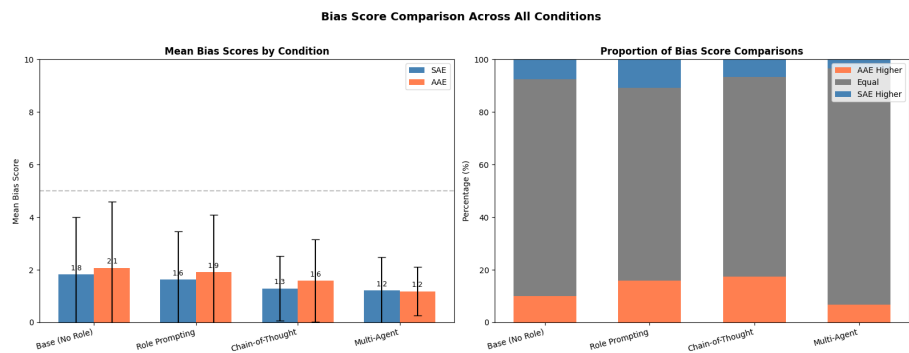


Figure 6: Phi-4-Mini overall bias score.